ZERO DIVISORS OF THE INTEGRAL SEDENIONS AND THE COMPACT G_2

PART IV OF THE OCTONION PROGRAM RECORD

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Abstract. We study the zero divisors of the integral sedenions, the Cayley–Dickson double of Coxeter's order of integral octonions, and their interaction with the arithmetic of the compact group G_2. Among the 57,600 sedenions whose two octonion halves are units, exactly 7,560 are zero divisors, namely those whose halves are imaginary and orthogonal; they fall into two $G_2(\mathbb{Z})$-orbits of sizes 6,048 and 1,512, distinguished by the type (Lipschitz or Hurwitz) of the quaternion subalgebra they generate, and the 84,672 ordered pairs (x, y) with $xy = 0$ form seven free $G_2(\mathbb{Z})$-orbits: the integral shadow of Moreno's theorem that the real unit zero-divisor pairs form the Lie group G_2. Over $\mathbb{R}$ the zero-divisor manifold is $G_2/\mathrm{Sp}(1)$ and its Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$ is computed in low weights. The conic of a quaternion subalgebra modulo p meets the maximal-parahoric class set of the compact G_2 at p in a pattern governed by the octahedral group S_4: for $p \in \{3, 5, 7, 11\}$ each conic lies in a single class determined by the type of the zero divisor, and for $p \geq 13$ it splits into S_4-orbits, with class stabilisers $192/m$ and $48/m$; this is proved for all $p \geq 5$. Zero-divisor periods of algebraic modular forms on G_2 at level 3 are defined; they are the D^1-periods of Gan and Savin's exceptional see-saw, and the period along the conic separates Iwahori-new forms from the old ones (through the degeneracy map), which are identified exactly as Gan–Gurevich-type lifts of the level-3 newforms of weights 6 and 10 with the classical form on D^1. Three constructions of a sedenion spectral geometry in the sense of Farnsworth and Boyle are shown to admit no finite Dirac operator beyond a scalar mass, and the Euler product of the norm-counting zeta function is shown to survive exactly up to the sedenions along the Cayley–Dickson ladder. All computations and code are deposited in the program's repository.

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1. Introduction and summary

Parts I and II of this record [1, 2] concern class sets and Hecke modules of the compact groups G_2 and F_4 built from Coxeter's integral octonions $\mathbb{O}_\mathbb{Z}$ (the E_8 lattice). The present part goes one rung further on the Cayley–Dickson ladder, to the sedenions $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$, and asks what the first failure of division has to do with the arithmetic developed so far. The answer is more structured than expected.

(a) The integral sedenions $\mathbb{S}_\mathbb{Z} = \mathbb{O}_\mathbb{Z} \oplus \mathbb{O}_\mathbb{Z}\ell$ form the lattice $E_8 \oplus E_8$. Their zero divisors with unit halves are *exactly* the elements $a + b\ell$ with a, b imaginary orthogonal units (Theorem 2.1); there are 7,560 of them, each with a 4-dimensional annihilator, and they form two $G_2(\mathbb{Z})$-orbits separated by the type of the quaternion subalgebra $\langle 1, a, b, ab\rangle$. The ordered zero-divisor pairs form seven free $G_2(\mathbb{Z})$-orbits (Theorem 2.2).

(b) Over $\mathbb{R}$ the unit zero divisors form $G_2/\mathrm{Sp}(1)$ and Moreno's G_2 is the sphere bundle of their annihilators. The Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$, a non-commutative algebra with continuous families of idempotents, has blocks of sizes $1, 3, 3, 6, 8, 10, 6$ on the representations of dimensions $1, 7, 14, 27, 64, 77, 77'$ (Section 3).

(c) The conic law (Theorem 4.4): the isotropic lines of $\mathrm{Im}\,\mathbb{H}_z \otimes \mathbb{F}_p$ form a conic $\cong \mathbb{P}^1(\mathbb{F}_p)$, and its intersection with the maximal-parahoric class set of the compact G_2 at p is governed by the octahedral group S_4. The two zero-divisor types index two classes for $p = 3, 5, 7, 11$ and the conics split into cube-orbits for $p \geq 13$.

(d) Zero-divisor periods of algebraic modular forms (Section 5) are the D^1-periods of the Gan–Savin see-saw [6]; the period along the conic vanishes exactly on the Iwahori-new forms of weight $2\omega_1$ at level 3, while their periods on other orbits do not.

(e) Negatives (Section 7): in the Farnsworth–Boyle framework of non-associative spectral geometry [8, 9] the sedenions admit, in all constructions tried, no finite Dirac operator beyond a scalar mass; the sedenion analogue of the Albert algebra is not Jordan; and the Euler product of the norm-counting zeta function dies one doubling past the sedenions, where Ramanujan's Δ first enters.

Everything in (a)–(d) was computed exactly (integer arithmetic, finite groups, or linear algebra at machine precision with independent checks); the conic law is proved for all $p \geq 5$ by a structural reduction to a finite computation (Section 4).

2. The integral sedenions and their zero divisors

Let $\mathbb{O}_\mathbb{Z}$ be Coxeter's maximal order of integral octonions, with norm N, trace tr and the 240 units. The Cayley–Dickson double $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ has product $(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c})$, conjugation $\overline{(a, b)} = (\bar{a}, -b)$ and norm $N(a, b) = N(a) + N(b)$; we take $\mathbb{S}_\mathbb{Z} = \mathbb{O}_\mathbb{Z} \oplus \mathbb{O}_\mathbb{Z}\ell$. On 200 random triples the laws behave as classically known: $\mathbb{S}$ is flexible and power-associative; it is not commutative, associative or alternative, and $N(xy)/(N(x)N(y))$ ranged over $[0.709, 1.316]$.

Theorem 2.1. *Let $x = (a, b) \in \mathbb{S}_\mathbb{Z}$ with $N(a) = N(b) = 1$. Then x is a zero divisor if and only if a and b are both imaginary ($\mathrm{tr} = 0$) and orthogonal. There are 7,560 such elements; each has a 4-dimensional annihilator $\{y : xy = 0\}$. No unit of $\mathbb{S}_\mathbb{Z}$ (norm 1) is a zero divisor.*

Proof. Exhaustive computation over all $240^2 = 57{,}600$ pairs (a, b) of units: the left-multiplication matrix L_x is singular exactly for the 7,560 pairs with $\operatorname{tr} a = \operatorname{tr} b = 0$ and $\langle a, b\rangle = 0$ (the other seven combinations of the three conditions never give a zero divisor), and its kernel has dimension 4 in every case. A unit has one half zero and all 230,400 products of units have norm 1. □

The real version (imaginary, orthogonal, equal norm) is due to Moreno and to Khalil–Yiu; the count and the dimension 4 reproduce Moreno's dimension count $11 + 3 = 14 = \dim G_2$ for the set of unit zero-divisor pairs [3].

Theorem 2.2. *The group $G_2(\mathbb{Z}) = \operatorname{Aut}(\mathbb{O}_\mathbb{Z})$ (order 12,096) acts on the 7,560 zero divisors with two orbits, of sizes 6,048 and 1,512, with stabilisers of orders 2 and 8. The quaternion subalgebra $\mathbb{H}_z = \langle 1, a, b, ab\rangle \cap \mathbb{O}_\mathbb{Z}$ contains 8 units (Lipschitz type) on the large orbit and 24 (Hurwitz type) on the small one, and these are exactly the numbers of zero divisors y in the annihilator of x among the 7,560. The $84{,}672 = 7 \cdot 12{,}096$ ordered pairs (x, y) of these zero divisors with $xy = 0$ form seven free $G_2(\mathbb{Z})$-orbits.*

Proof. Computation with the 12,096 automorphisms in the Coxeter basis (each verified to be a sedenion automorphism acting diagonally on the two halves); the orbit and stabiliser counts are exact. The map $(a, b) \mapsto (a, -b)$ is an automorphism of $\mathbb{S}$; the half-swap preserves the zero-divisor set but is not an automorphism. □

Remark 2.3. Moreno's theorem says the real unit zero-divisor pairs form a space homeomorphic to G_2; Theorem 2.2 is its integral shadow: seven copies of $G_2(\mathbb{Z})$, acted on freely.

3. The zero-divisor manifold and its Hecke algebra

Over $\mathbb{R}$, a unit zero divisor is a pair (a, b) of orthonormal imaginary octonions. G_2 acts transitively on such pairs, so the zero-divisor manifold is $ZD = G_2/K$ with K the stabiliser of one pair.

Proposition 3.1. *K is 3-dimensional with Lie algebra $\mathfrak{su}(2)$; it fixes $\mathbb{H}_z$ pointwise and acts on the 4-dimensional annihilator of $z = (a, b)$ as $\mathrm{Sp}(1)$ acts on $\mathbb{H}$ (the commutant of K on the annihilator is 4-dimensional). Hence $ZD = G_2/\mathrm{Sp}(1)$ is the Stiefel manifold of orthonormal pairs in $\mathbb{R}^7$, of dimension 11, and Moreno's G_2 is the unit-sphere bundle of the annihilators over it.*

Proof. Computed from the 14 derivations of $\mathbb{O}$ in an orthonormal basis: the derivations killing a and b form a 3-dimensional subalgebra, which kills $\langle 1, a, b, ab\rangle$ and preserves the annihilator, on which its commutant has dimension 4. □

In Gan–Savin's notation [6], $K = D^1$ is the group of norm-one elements of the quaternion subalgebra $D = \mathbb{H}_z$, acting by $g \cdot (a, b) = (a, b\bar{g})$; it is the pointwise stabiliser of D in $\operatorname{Aut}(\mathbb{O})$.

The Hecke algebra of the homogeneous space is $\mathcal{H}(G_2, K) = \bigoplus_\pi M_{m_\pi}(\mathbb{C})$ with $m_\pi = \dim \pi^K$ (Frobenius reciprocity: m_π is the multiplicity of π in $L^2(ZD)$). Decomposing $7 \otimes 7$, $7 \otimes 14$, $\mathrm{Sym}^3 7$ and $\mathrm{Sym}^2 14$ by the Casimir and counting K-fixed vectors:

π	1	7	14	27	64	77	77′
$m_\pi = \dim \pi^K$	1	3	3	6	8	10	6

The 7 gives 3 because K fixes exactly $\operatorname{Im}\mathbb{H}_z \subset \operatorname{Im}\mathbb{O}$; the 14 gives 3 because the centraliser of K in $\mathfrak{g}_2$ is the other $\mathfrak{su}(2)$ of $\mathfrak{so}(4) \subset \mathfrak{g}_2$. Since (G_2, K) is not a Gelfand pair, the idempotents of each block form continuous families (Grassmannians). The representation theory here is classical; the identification of ZD with the sedenion zero divisors is what makes it ours. Every weight in the Path A table of compact G_2 forms (Part II) has $m_\pi > 0$, which is what makes the periods of Section 5 possible.

4. Quaternion subalgebras and the conic law

4.1. **Quaternion subalgebras of the Coxeter order.** Every zero divisor $z = (a, b)$ determines $\mathbb{H}_z = \langle 1, a, b, ab\rangle \cap \mathbb{O}_{\mathbb{Z}}$, a quaternion order of the definite quaternion algebra it spans. Counting ordered orthogonal imaginary unit pairs per subalgebra ($6 \cdot 4 = 24$ in both cases) gives:

Proposition 4.1. *$\mathbb{O}_{\mathbb{Z}}$ contains 63 quaternion subalgebras of Hurwitz type (24 units) and 252 of Lipschitz type (8 units). Their stabilisers in $G_2(\mathbb{Z})$ have orders 192 and 48; the pointwise stabilisers have orders 8 and 2 (the stabilisers of the zero divisors of Theorem 2.2), and in both cases the quotient acting faithfully on $\operatorname{Im}\mathbb{H}_z$ is the octahedral group S_4.*

Proof. $1512/24 = 63$, $6048/24 = 252$; $12096/63 = 192$, $12096/252 = 48$; the pointwise stabilisers were computed directly, and $192/8 = 48/2 = 24 = |S_4|$, with S_4 the full rotation group of the cube $\{\pm i, \pm j, \pm k\}$, realised in both cases by the computation of Section 5 (orbit sizes $6, 8, 12$ below). □

Proposition 4.2. *The imaginary integral units are the 126 roots of the E_7 lattice $\operatorname{Im}\mathbb{O}_{\mathbb{Z}}$, and the 7,560 zero divisors of Theorem 2.1 are exactly the ordered pairs of orthogonal E_7 roots (60 roots are orthogonal to any given root, a D_6). The zero-divisor mass $7560/12096 = 5/8$ is thus a Siegel representation number of the binary form $2I_2$ by E_7 divided by $|G_2(\mathbb{Z})|$, and the zero divisors of each norm are governed by Siegel's local–global formula for representations by E_7, even though their counting series has no Euler product of its own.*

Proposition 4.3. *For a zero-divisor pair $x = (a, b)$, $y = (c, d)$ with $xy = 0$ one has $d = \pm(ac)b$, so the pair is the triple (a, b, c) with c a unit perpendicular to $\mathbb{H}_x$. The seven free $G_2(\mathbb{Z})$-orbits of Theorem 2.2 are separated by the types (Hurwitz 24 or Lipschitz 8) of the five quaternion subalgebras $\langle a, b\rangle, \langle c, d\rangle, \langle a, c\rangle, \langle b, c\rangle, \langle ab, c\rangle$: the seven realised type-patterns are $(8, 8, 8, 24, 8)$, $(8, 8, 24, 8, 8)$, $(8, 24, 8, 8, 24)$, $(8, 24, 24, 24, 24)$ for Lipschitz x and $(24, 8, 8, 24, 8)$, $(24, 8, 24, 8, 8)$, $(24, 24, 8, 8, 24)$ for Hurwitz x, with equal counts within each type as the freeness requires.*

4.2. **The conic.** Fix an odd prime p. The space $\operatorname{Im}\mathbb{O} \otimes \mathbb{F}_p$ is a nondegenerate 7-dimensional quadratic space; its isotropic lines number $(p^6-1)/(p-1)$ and form the maximal-parahoric level-p structures of the compact G_2, so that the class set is $X_p = G_2(\mathbb{Z})\backslash\{\text{isotropic lines}\}$ (Part I). For a zero divisor z, $\operatorname{Im}\mathbb{H}_z \otimes \mathbb{F}_p$ is a nondegenerate ternary space, so its isotropic lines form a conic $C_z \cong \mathbb{P}^1(\mathbb{F}_p)$ with $p+1$ points, on which $\operatorname{Stab}(\mathbb{H}_z)$ acts through $S_4 \subset \mathrm{PGL}_2(\mathbb{F}_p)$.

Theorem 4.4 (Conic law)**.** *Let $p \geq 5$ and z a zero divisor. The $G_2(\mathbb{Z})$-classes of the points of C_z are exactly the S_4-orbits on $\mathbb{P}^1(\mathbb{F}_p)$, whose sizes lie in $\{6, 8, 12, 24\}$ (cube faces, vertices, edges, and generic), and the stabiliser in $G_2(\mathbb{Z})$ of a conic point in an orbit of size m has order $192/m$ (Hurwitz type) or $48/m$ (Lipschitz type). The number of classes a conic meets is the number of S_4-orbits on $\mathbb{P}^1(\mathbb{F}_p)$, which by Burnside's lemma is*

$$\frac{1}{24}\Big((p{+}1){+}3f_{-1}{+}6f_{-2}{+}8f_3{+}6f_4\Big), \qquad f_{-1} = 2 \text{ if } \left(\tfrac{-1}{p}\right) = 1,\ f_{-2} = 2 \text{ if } \left(\tfrac{-2}{p}\right) = 1,\ f_3 = 2 \text{ if } p \equiv 1\ (3),\ f_4 = 2 \text{ if}$$

and 0 otherwise (the face, edge and vertex rotations of the cube have 2 or 0 fixed points on the conic according to these conditions). Equivalently, the conic carries a 6-point orbit (stabilisers 8 and 32) exactly when $p \equiv 1\ (4)$, an 8-point orbit (6 and 24) exactly when $p \equiv 1\ (3)$, a 12-point orbit (4 and 16) exactly when $(\frac{-2}{p}) = 1$, and all remaining points lie in 24-point orbits (2 and 8); this was verified directly for all $p \leq 41$. The orbit structure is thus Eisenstein data of level 24: the number of classes a conic meets is $\frac{p+1}{24} + \frac{3}{4}[p \equiv 1\,(4)] + \frac{1}{2}[(\frac{-2}{p}) = 1] + \frac{2}{3}[p \equiv 1\,(3)]$, and its Dirichlet series over p is a combination of $\zeta(s-1)$, $\zeta(s)$ and the L-functions of the characters of conductors 3, 4 and 8, while the cuspidal content of the zero divisors lies entirely in the periods

of Section 5. In particular each conic lies in a single class exactly for $p \in \{5, 7, 11, 23\}$ *(*$p = 23$ *is the regular orbit,* 24 *points with stabilisers* 2 *and* 8*), and for* $p \geq 29$ *never; for* $p = 3$ *the conic (*4 *points) also lies in a single class, with the exceptional stabilisers* 432 *and* 36*. The count depends only on* p *modulo* 24*.*

Lemma 4.5. *For* $p \geq 5$*, an isotropic line lying on the conic of a quaternion subalgebra of given type lies on no other conic of that type.*

Proof. Let $\mathbb{H}_1 \neq \mathbb{H}_2$ be suborders of the same type, $M_i = \operatorname{Im} \mathbb{H}_i$ (rank 3). Since $\mathbb{H}_i = B_i \cap \mathbb{O}_\mathbb{Z}$ for the $\mathbb{Q}$-subalgebra B_i it spans, $B_1 \neq B_2$, and two distinct quaternion subalgebras of $\mathbb{O}$ meet in a subalgebra of dimension ≤ 2 (a 3-dimensional subalgebra of a quaternion algebra does not exist); hence $M_1 \cap M_2$ has rank ≤ 1. The conics C_1, C_2 modulo p share a point only if $\bar{M}_1 \cap \bar{M}_2 \subset \operatorname{Im} \mathbb{O} \otimes \mathbb{F}_p$ contains a nonzero isotropic vector. Now $\dim_{\mathbb{F}_p}(\bar{M}_1 \cap \bar{M}_2) = 6 - \operatorname{rank}_p(M_1 + M_2)$, and $\operatorname{rank}_p(M_1 + M_2) = \operatorname{rank}(M_1 + M_2)$ unless p divides the index of $M_1 + M_2$ in its saturation. So either (i) p divides that index, or (ii) $\bar{M}_1 \cap \bar{M}_2$ is the reduction of $M_1 \cap M_2$, which is 0 or $\mathbb{Z}v$ with v a common imaginary vector, isotropic mod p only if $p \mid N(v)$.

Computation over all $252 \cdot 251/2 = 31{,}626$ Lipschitz pairs and $63 \cdot 62/2 = 1{,}953$ Hurwitz pairs (code `g2table/lemma43.py`): the saturation indices take the values $1, 2, 3, 4, 5, 6, 8, 10$ (Lipschitz) and $1, 2, 3, 4, 5, 6, 7, 8$ (Hurwitz), so their prime factors lie in $\{2, 3, 5, 7\}$; the pairs with $\operatorname{rank}(M_1 + M_2) = 5$ (6,426 Lipschitz, 1,197 Hurwitz) have common directions of norm 1 or 2 (Lipschitz) and 1 or 3 (Hurwitz). Hence for $p \geq 11$ case (i) never occurs and case (ii) gives no isotropic vector: the conics are disjoint. For $p = 5$ and $p = 7$, case (i) occurs for $15 + 75$ pairs at $p = 5$ and 23 Hurwitz pairs at $p = 7$; for each of these the intersection $\bar{M}_1 \cap \bar{M}_2$ was computed over $\mathbb{F}_p$ and found to contain no isotropic vector, while case (ii) is excluded as before since $p \nmid N(v) \in \{1, 2, 3\}$. This proves the lemma for all $p \geq 5$. (For $p = 3$ it fails: $3 \mid N(v)$ for the Hurwitz pairs with a common direction of norm 3, and the conics do share points, which is why the stabilisers $432, 36$ at $p = 3$ exceed $192/4$ and $48/4$.) □

Proof of Theorem 4.4 from Lemma 4.5. Let $\ell \in C_z$ and $g \in G_2(\mathbb{Z})$ with $g\ell \in C_z$. Then $g\ell$ lies on the conics of $\mathbb{H}_z$ and of $g\mathbb{H}_z$, both of the type of z; by the lemma $g\mathbb{H}_z = \mathbb{H}_z$, so $g \in \operatorname{Stab}(\mathbb{H}_z)$. Hence the $G_2(\mathbb{Z})$-orbit of ℓ meets C_z exactly in its $\operatorname{Stab}(\mathbb{H}_z)$-orbit, which is its S_4-orbit since the pointwise stabiliser acts trivially on C_z, and the stabiliser of ℓ in $G_2(\mathbb{Z})$ equals its stabiliser in $\operatorname{Stab}(\mathbb{H}_z)$, of order $|\operatorname{Stab}(\mathbb{H}_z)|/m$. The orbit sizes of $S_4 \subset \mathrm{PGL}_2(\mathbb{F}_p)$ on $\mathbb{P}^1(\mathbb{F}_p)$ are $24/|\text{point stabiliser}|$ with point stabilisers the cyclic subgroups of orders $4, 3, 2, 1$, giving $6, 8, 12, 24$; S_4 is transitive on $\mathbb{P}^1(\mathbb{F}_p)$ iff $p + 1 \in \{6, 8, 12\}$ (the case 4 being $p = 3$, where the lemma fails and the stabilisers are larger). □

Computation 4.6. Independently of the proof, exact orbit computations for $p = 3, 5, 7, 11, 13, 17, 19$ (four to forty zero divisors of each type per prime) give the class structure of the conic points shown in Table 1; the stabiliser identity $|\operatorname{Stab}(\ell)| = |\operatorname{Stab}(\mathbb{H}_z)|/m$ holds in every case with $p \geq 5$, as the theorem predicts. The classes touched were also matched against the full class sets X_5 (3 classes, stabilisers $32, 8, 6$: the 6-class is touched by no zero divisor) and X_7 (8 classes).

Remark 4.7. The incidence counts are now explained: $63(p + 1)$ Hurwitz and $252(p + 1)$ Lipschitz incidences fall on classes of sizes $12096\,m/192$ and $12096\,m/48$, one conic through each point.

5. Zero-divisor periods

5.1. **Definition.** Let f be an algebraic modular form of weight π for the compact G_2 at a parahoric level at $p = 3$: a function on the level-3 structures x (isotropic lines, planes or flags

p	$p+1$	Lipschitz: (points, stabiliser)	Hurwitz: (points, stabiliser)
3	4	$(4, 36)$	$(4, 432)$
5	6	$(6, 8)$	$(6, 32)$
7	8	$(8, 6)$	$(8, 24)$
11	12	$(12, 4)$	$(12, 16)$
13	14	$(8, 6) + (6, 8)$	$(8, 24) + (6, 32)$
17	18	$(12, 4) + (6, 8)$	$(12, 16) + (6, 32)$
19	20	$(12, 4) + (8, 6)$	$(12, 16) + (8, 24)$

TABLE 1. Class structure of the conic points of a zero divisor; stabilisers are $48/m$ and $192/m$ for $p \geq 5$.

mod 3) with $f(x\gamma) = \pi(\gamma)^{-1} f(x)$, in the conventions of [2]. For a zero divisor z with continuous stabiliser K_z, and $u \in \pi^{K_z}$, put

$$P_{z,u}(f) = \sum_{x \text{ compatible with } z} \langle f(x), u \rangle, \qquad x \text{ compatible} \iff \text{the line of } x \text{ lies in } \operatorname{Im} \mathbb{H}_z \bmod 3,$$

and more generally, for any orbit O of the finite group $\mathrm{Stab}(\mathbb{H}_z)$ on the structures, $P_{z,O}(f) = \mathrm{proj}_{\pi^{K_z}} \sum_{x \in O} f(x)$. Every zero divisor has exactly 4 compatible lines (the conic at $p = 3$).

5.2. **Trivial weight.** At the maximal parahoric level X_3 has two classes (stabilisers $432, 36$) and T_2-eigenvalues 126 (Eisenstein) and 9 (the lift of $3.6.a.a$). By Theorem 4.4 the four conic points of a Hurwitz zero divisor lie in the 432-class and those of a Lipschitz one in the 36-class, so $P_z(f) = 4f(\text{class of } z)$: for the cusp form, normalised $f = (-1, 1/12)$, the periods are -4 and $1/3$.

5.3. **Weights 14 and 27.** Weight 14 (adjoint): one form at level 3 ($T_2 = -18$, at the plane and flag levels), whose conic periods vanish for both types. Weight 27 ($2\omega_1$): at the Iwahori level a 7-dimensional space with $T_2 \in \{40.5\ (\times 2),\ 0\ (\times 3),\ -10.768,\ 6.268\}$, the irrational pair being the roots of $x^2 + 4.5x - 67.5$ and new at the Iwahori level; the others descend to the maximal parahoric levels, where $(T_2, T_5) = (81/2,\ 2034/5)$ and $(0,\ 11466/25)$ (twice), computed with the calibration $(9, 810)$ and $(126, 19530)$ in trivial weight.

Computation 5.1. (i) The conic period $P_{z,u}$ vanishes identically on the Iwahori-new forms and is nonzero on all old forms, for both types. (ii) Over all $\mathrm{Stab}(\mathbb{H}_z)$-orbits (51 orbits for Lipschitz, 21 for Hurwitz type), the Iwahori-new forms have nonzero $P_{z,O}$ on 41/51 resp. 16/21 orbits, vanishing exactly on the orbits containing the conic flags (and a few others); the old forms are nonzero on every orbit. (iii) For the Iwahori-new forms the sum of f over the four flags above *each* conic point vanishes separately, before any projection.

Remark 5.2 (what the conic vanishing is)**.** Item (iii) explains item (i) completely: the sum of f over the planes containing a fixed line is the degeneracy map from the Iwahori level to the maximal-parahoric line level, and Iwahori-new forms lie in its kernel for *every* line. The conic period factors through this degeneracy map, so its vanishing on new forms is the definition of newness, not a property of the zero divisors; the Hurwitz/Lipschitz type and the K-projection play no role in it. The content of zero-divisor periods is therefore (a) at the maximal parahoric level, where the conic law (Theorem 4.4) governs them, and (b) in orbit-periods $P_{z,O}$ that do not factor through degeneracy maps, which is where the new forms are nonzero, in accordance with the global expectation of the next subsection.

5.4. **The maximal parahoric level 7 and the selection rule.** At the maximal parahoric level $p = 7$ the class set has 8 classes (stabilisers $2, 18, 42, 6, 6, 2, 6, 24$), and the conics of the Hurwitz and Lipschitz zero divisors lie in the 24-class and in one 6-class (Theorem 4.4). Since the $p+1$ conic points form a single orbit of the finite group $\mathrm{Stab}(\mathbb{H}_z)$ when $p \le 11$, the conic sum $\sum_{\ell \in C_z} f(\ell)$ is the projection of $f(\ell_0)$ onto the $\mathrm{Stab}(\mathbb{H}_z)$-fixed vectors of V_π, which gives a selection rule: $\dim V_\pi^{\mathrm{Stab}(\mathbb{H}_z)}$ is 1 for the trivial weight, 0 for the 7 and the 14, and 1 (Hurwitz) or 2 (Lipschitz) for the 27. The rule for all weights up to 77, computed from the characters of $\mathrm{Stab}(\mathbb{H}_z)$ on the eigenvalues of its elements on the 7, is

weight	1	7	14	27	64	77	77′
$\dim V_\pi^{\mathrm{Stab}(\mathbb{H}_z)}$, Lipschitz ($\lvert\,\mathrm{Stab}\,\rvert = 48$)	1	0	0	2	1	1	3
$\dim V_\pi^{\mathrm{Stab}(\mathbb{H}_z)}$, Hurwitz ($\lvert\,\mathrm{Stab}\,\rvert = 192$)	1	0	0	1	0	0	2

so from weight 64 on the two types differ: in weights 64 and 77 every form is dark on Hurwitz conics and (generically) lit on Lipschitz ones. This was verified in weight 64 at level 7 (97 forms, the 64 cut out of $7 \otimes 14$ by the character projector over $G_2(\mathbb{Z})$): all 97 conic sums vanish for the Hurwitz type and none for the Lipschitz type. Accordingly all conic sums vanish identically in weights 7 and 14 for every form, and in weight 27 (50 forms at this level) the periods are nonzero for every form and both types; in trivial weight (7 tempered cusp forms) likewise. So zero-divisor periods do not vanish on tempered forms: they are generic nonzero functionals, in accordance with the nonvanishing predicted by the see-saw, and their content lies in the conic law and the selection rule rather than in any vanishing.

The same computation extends Theorem 5.3 to level 7: the five newforms of $S_{10}(\Gamma_0(7))$ appear in weight $2\omega_1$ at the maximal parahoric level with exactly the predicted (T_2, T_3), namely $(58.351, 116.103)$, $(37.021, 188.629)$, $(35.169, 28.970)$ (multiplicity 2), $(14.331, 164.808)$ (multiplicity 2) and $(1.378, 80.601)$; the first two violate the tempered bound, the others pass it. In trivial weight the level-7 lifts of weight-6 newforms have no fixed vector at this parahoric and live at the plane parahoric.

5.5. **Interpretation.** In [6] the pair $D^1 \subset \mathrm{Aut}(\mathbb{O})$ sits in a see-saw inside the adjoint E_7 of the Albert algebra, with $\mathrm{Aut}(\mathbb{O}) \times \mathrm{PGSp}_6$ and $D^1 \times G_D$ (G_D of type D_6) as dual pairs; the theta lift of the trivial representation of D^1 is a residue of a Siegel Eisenstein series on G_D, and by Pollack's integral and Chenevier's identity $L(s, \pi, \mathrm{Spin}) = \zeta(s) L(s, \pi', \mathrm{Std})$ for G_2-parameters, the D^1-period of a form on the compact G_2 is governed by $L(1, \pi', \mathrm{Std}) \neq 0$. Our periods are D^1-periods with specific local data at 3. Computation 5.1(ii) is in accordance: the genuine (Iwahori-new) forms have nonzero D^1-periods on orbits that do not factor through degeneracy maps; the vanishing along the conic is the degeneracy map itself (Remark above). The old forms are identified. Writing $\mathrm{SO}(4) = (\mathrm{SL}_2^{\mathrm{long}} \times \mathrm{SL}_2^{\mathrm{short}})/\mu_2 \subset G_2$ for the stabiliser of a quaternion subalgebra, with $7 = (2,2) \oplus (1,3)$ and $\mathrm{SL}_2^{\mathrm{long}} = D^1$, consider the Arthur-type parameter $\psi_f = \varphi_f \boxtimes \nu$ with a classical newform f of weight k on the long-root SL_2 and the Arthur SL_2 on the short root. Since the Hecke eigenvalue of an algebraic modular form is the Satake transform of the unitary local representation (the weight enters only at infinity), $T_p = p^3 \chi_7(s_p) - 1$ in every weight, and here

$$\chi_7(s_p) = \left(p + 1 + \tfrac{1}{p}\right) + \left(p^{1/2} + p^{-1/2}\right) \frac{a_p(f)}{p^{(k-1)/2}}, \qquad T_p = \frac{p^3(p^2+p+1)}{p} - 1 + \frac{(p+1)\,a_p(f)\,p^3}{p^{k/2}}.$$

Theorem 5.3. *At level* 3 *the trivial-weight cusp form* $(T_2, T_5, T_7) = (9, 810, 2472)$ *is* $\psi_{3.6.a.a}$ *$(a_2, a_5, a_7 = -6, 6, -40)$, and the weight-$2\omega_1$ old forms are* $\psi_{3.10.a.a}$ *$(a_2, a_5 = 18, -1530$; $T_2 = 81/2$, $T_5 = 2034/5)$ and* $\psi_{3.10.a.b}$ *$(a_2, a_5 = -36, -1314$; $T_2 = 0$, $T_5 = 11466/25$, multiplicity 2 at*

the maximal parahoric level), with the weight rule $k = 6 + 2a$ for G_2-weight $a\omega_1$. All six (resp. four) Hecke eigenvalues agree exactly; the prime 7 was not used in finding the dictionary.

These are the Gan–Gurevich non-tempered forms [5] seen on the compact G_2: the classical form lives on D^1, which is why their D^1-periods are large, and the two weight-10 forms have opposite Atkin–Lehner signs at 3 ($a_3 = \pm 81$), a natural source of the multiplicity difference (to be confirmed against Arthur's multiplicity formula). No level-one forms exist in weights 14, 27, $77'$ ($G_2(\mathbb{Z})$-invariants vanish). The Iwahori-new irrational pair is not of this type and is a genuine (tempered) form.

6. The annihilator lattices, triality, and the next rung

Proposition 6.1 (annihilator lattices)**.** *Let $x = (a, b)$ be a zero divisor of the integral sedenions with unit halves and $\mathbb{H}_x = \langle 1, a, b, ab\rangle$ its quaternion subalgebra. The orthogonal complement $\mathbb{H}_x^{\perp} \cap \mathbb{O}_{\mathbb{Z}}$, whose units are exactly the partners c with $(c, \pm(ac)b)$ a zero divisor annihilating x, is the root lattice D_4 (determinant 4, 24 roots) when x is of Hurwitz type and the cubic lattice $\mathbb{Z}^4$ (8 units) when x is of Lipschitz type. Every imaginary unit lies in exactly 3 Hurwitz and 12 Lipschitz quaternion subalgebras, which gives the $12 + 48$ partners of a unit.*

Proposition 6.2 (the seven orbits, explained)**.** *For a Hurwitz zero divisor the 24 partners split as $8+8+8$ according to which of the three axes a, b, ab of the frame the partner c completes to a Hurwitz subalgebra; each partner completes exactly one axis, never two. These three classes are the three triality classes of the roots of D_4, and they are the three orbits of pairs with Hurwitz x. For a Lipschitz zero divisor the 8 partners are the $\pm$ unit vectors of $\mathbb{Z}^4$: three of the four axes carry the partners completing one frame axis each and the fourth carries the two partners completing all three, the pattern $(8, 24, 24, 24, 24)$. The missing eighth pattern $(24, 24, 24, 24, 24)$, an all-Hurwitz pair, cannot occur.*

Theorem 6.3 (four rungs of the Cayley–Dickson ladder)**.** *Let A_{2^k} ($k = 4, 5, 6, 7$) be the integral sedenions, 32-ions, 64-ions and 128-ions, the iterated Cayley–Dickson doubles of $\mathbb{O}_{\mathbb{Z}}$, with $m = 2^{k-3}$ octonion slots, and call an element $(x, y) \in A_{2^k} = A_{2^{k-1}} \oplus A_{2^{k-1}}$ unit-half if x and y are units of $A_{2^{k-1}}$ (the $240\,m/2$ elements with a root of E_8 in one slot). Then (x, y) is a zero divisor if and only if x and y are imaginary and perpendicular. The numbers of unit-half zero divisors are*

$$7{,}560, \qquad 98{,}280, \qquad 625{,}320, \qquad 3{,}061{,}800,$$

each equal to the number of ordered pairs of perpendicular imaginary units of the previous rung ($126 + 240(m/2 - 1)$ imaginary units; the pairs within the first slot are the 7,560 sedenion zero divisors, those within any other slot the 30,240 perpendicular pairs of E_8 roots, and any two different slots are perpendicular). The annihilator of a unit-half zero divisor has dimension $4 + 8t$ with $0 \le t \le m - 1$: $\{4\}$ at the sedenions, $\{4, 12\}$ at the 32-ions (59,472 and 38,808), $\{4, 12, 20, 28\}$ at the 64-ions (302,064, 171,360, 46,368, 105,528), and $\{4, 12, \ldots, 60\}$ at the 128-ions (1,077,936, 694,176, 290,304, 395,136, 76,608, 108,864, 172,704, 246,072), exactly the eight values $4 + 8t$ predicted before the computation. The strata are determined by the slots: two units in the same slot give the maximal dimension; two slots of one sedenion copy give 12; slots in different copies give 4, except that the 7,560 sedenion zero-divisor pairs placed in the two outermost slots give 20, and in the two innermost slots the pairs of distinct imaginary units give 20 ($126 \cdot 126 - 252$) and the coincident ones 28; the elements lying in a quaternion-sized subalgebra with a new unit ($b = \pm a$ or $b = \pm 1$) always reach the maximal dimension. In particular the annihilator of a sedenion zero divisor grows from 4 to 12 to 28 to 60 up the ladder. The automorphism group of the integral 2^k-ions, $G_2(\mathbb{Z}) \times (\mathbb{Z}/2)^{k-3}$, has 2, 24 and 170 orbits on the unit-half zero divisors at $k = 4, 5, 6$, with stabilisers of orders $8, 32$; $6, 8, 16, 32, 192$;

$12, 16, 32, 64, 384, 432$ *and* 24,192 *(the six orbits of size* 4 *at the* 64*-ions are the real unit placed in two different slots).*

Proposition 6.4 (annihilator lattices up the ladder)**.** *The integral annihilator of a unit-half zero divisor, as a lattice of norm-*2 *vectors, is a root lattice, and the root systems are: at the sedenions* D_4 *(Hurwitz) and* A_1^4 *(Lipschitz); at the* 32*-ions* $D_4 \oplus E_8$ *and* $A_1^4 \oplus E_8$ *for the sedenion zero divisors, and* $D_6 \oplus D_6$ *for the perpendicular root pairs of the second copy and for the diagonal mixed elements; at the* 64*-ions* $D_4 \oplus E_8^3$ *for a same-slot Hurwitz element,* $D_6^2 \oplus D_4^2$ *for the sedenion pairs in the outermost slots and* $D_4^4 \oplus A_1^4$ *for the inner cross pairs of distinct imaginary units (the two* 20*-dimensional strata). The step of* 8 *in* $4 + 8t$ *is one copy of* E_8 *per octonion slot, and* D_6 *is the root system of* E_8 *perpendicular to a perpendicular pair of its roots.*

Relation to the literature. Over the reals the sedenion statement is classical: Moreno proved that zero divisors of the Cayley–Dickson algebras are imaginary with annihilators of dimension divisible by 4, and Biss, Dugger and Isaksen that (a, b) is a zero divisor exactly when a, b are imaginary, perpendicular and of equal norm, with $\dim \operatorname{ann} \leq 2^n - 4n + 4$ (G. Moreno, *The zero divisors of the Cayley–Dickson algebras over the real numbers*, Bol. Soc. Mat. Mex. 1998, and *Constructing zero divisors in the higher dimensional Cayley–Dickson algebras*, 2005; D. Biss, D. Dugger, D. Isaksen, *Large annihilators in Cayley–Dickson algebras*, Comm. Algebra 2008). Our dimensions $4, 12, 20, 28$ satisfy both constraints. What is new here is integral: the lattices D_4, A_1^4, their growth by E_8, the explanation of the seven orbits through the triality of D_4 and the axes of $\mathbb{Z}^4$, the exact unit-half census at the 32- and 64-ions, and the conic law and selection rule of the earlier sections.

Theorem 6.5 (structure of the annihilator, and the law at every rung)**.** *Let* x, y *be pure-slot units of* A_n *(an octonion unit in one slot of the standard basis),* $X = (x, y) \in A_{2n}$*, and* $Q = (z, w)$ *arbitrary. Then*

- *(i)* $N(XQ) = 2N(Q) - 2\langle z, (\bar{x}, \bar{w}, y)\rangle$*, where* $(a, b, c) = a(bc) - (ab)c$ *is the associator. Consequently* X *is a zero divisor if and only if some* $w \neq 0$ *is anti-associative with* $\bar{x}$ *and* y*, i.e.* $\bar{x}(\bar{w}y) + (\bar{x}\bar{w})y = 0$*, and then* $\operatorname{ann}(X) = \{(\bar{x}(\bar{w}y),\, w)\}$ *over the kernel of the anti-associator* $\Phi_{x,y} \colon w \mapsto \bar{x}(\bar{w}y) + (\bar{x}\bar{w})y$*.*
- *(ii)* $\Phi_{x,y}$ *maps each slot of* A_n *into a single slot (standard units multiply by symmetric difference of their index sets, up to sign), so* $\ker \Phi_{x,y}$ *is the direct sum of its single-slot kernels and* $\dim \operatorname{ann}(X) = \sum_R \dim \ker(\Phi_{x,y}|_{\text{slot } R})$*.*
- *(iii) Each single-slot restriction involves only* $\mathbb{O}$ *and at most three standard units, hence lives in a Cayley–Dickson subalgebra of dimension at most* 64*. The unit-half law and the stratification* $4 + 8t$ *therefore hold at every rung as soon as they hold for the* 64*-ions, which Theorem 6.3 establishes (the* 128*-ions serving as an independent check).*

Proof sketch. (i) Left and right multiplication by a pure-slot element u multiply norms: $N(uq) = N(u)N(q) = N(qu)$, by induction up the ladder from the octonions, since for $u = (u', 0)$ one has $(u', 0)(z, w) = (u'z, wu')$ and for $u = (0, u')$ one has $(0, u')(z, w) = (-\bar{w}u', u'\bar{z})$, and similarly on the right. Expanding $N(XQ) = N(xz - \bar{w}y) + N(wx + y\bar{z})$ with this and the adjoint identities $\langle ab, c\rangle = \langle b, \bar{a}c\rangle = \langle a, c\bar{b}\rangle$ of every Cayley–Dickson algebra gives (i); the equality case of Cauchy–Schwarz gives the anti-associativity criterion and the formula for the annihilator. (ii) is the slot bookkeeping of the standard basis. (iii) follows. The identity (i) and the slot decomposition (ii) were also verified exactly on all 230,400 unit-half pairs of the 32-ions, where the single-slot kernel dimensions sum to the annihilator dimensions 4 and 12 in exactly 59,472 and 38,808 cases. □

Remark 6.6. By Theorem 6.5, the unit-half law and the stratification $4+8t$ hold at every rung of the integral Cayley–Dickson ladder, modulo the standard facts about the standard basis used in (ii) and (iii); the stratum t counts the slots R whose anti-associator block is singular.

The law thus persists through the rung (32-ions) where the norm-counting Euler product is lost ($E_4^4 = E_{16} + \frac{3456000}{3617} E_4\Delta$): multiplicativity and the zero-divisor geometry are independent phenomena.

Computations: `zd32.py` and `zd64.py` in the repository (left-multiplication matrices of the 32- and 64-ions on the 230,400 and 921,600 unit-half candidates; annihilator dimensions by rank; the law verified anchor by anchor at all three rungs).

7. Negatives and boundaries

7.1. **Sedenion spectral geometries.** Farnsworth and Boyle [8, 9] extend Connes' spectral triples to non-associative algebras: the Hilbert space must be a bimodule in the algebra's own class (every identity surviving the replacement of one algebra element by a vector), the order conditions become vanishing associators in $B = \Omega A \oplus H$, and the gauge group is $\mathrm{Aut}(A)$. For the sedenions acting on themselves: the flexible order-zero law holds and the alternative ones fail; the derivation algebra is 14-dimensional and inner (inside $\mathfrak{so}(16) \oplus \mathbb{R}$); the fermions decompose as $2(1+7)$ under G_2 with the S_3 of $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ as a family symmetry; the commutant of the left action is trivial, forcing $\gamma = \pm 1$ and $D_F = 0$. On two copies $\mathbb{S} \oplus \mathbb{S}$ with $\gamma = \mathrm{diag}(1, -1)$, the flexible order-one condition forces D_F to be a multiple of the identity for either real structure; with the opposite bimodule on the second copy, no D_F survives; on the complex sedenions $\mathbb{C} \otimes \mathbb{S}$ with real-linear D_F, only complex scalars survive. The sedenion analogue $H_3(\mathbb{S})$ of the Albert algebra violates the Jordan identity (relative size 0.96; $H_3(\mathbb{O})$ satisfies it to 10^{-15}), so the Jordan-geometry route [9] excludes the sedenions and admits the Albert algebra. In every construction the zero divisors do not enter the action.

7.2. **Euler products along the ladder.** The norm-counting Dirichlet series $\sum r(n) n^{-s}$ has an Euler product iff r is multiplicative. It does for $\mathbb{Z}$, $\mathbb{Z}[i]$, the Hurwitz quaternions (D_4), the Coxeter octonions (E_8, $r = 240\sigma_3$) and the integral sedenions ($E_8 \oplus E_8$, $r = 480\sigma_7$) — the last only because $M_8(\mathrm{SL}_2(\mathbb{Z}))$ has no cusp forms — and fails for the integral 32-ions (E_8^4), since $E_4^4 = E_{16} + \frac{3456000}{3617} E_4\Delta$. The counting zeta function of the 32-ions, $\frac{16320}{3617}\zeta(s)\zeta(s-15) + c\, L(E_4\Delta, s)$, has two zeros on its critical line $\Re s = 8$ below $t \approx 15$ and none up to $t = 100$, where the Eisenstein part dominates by 10^5–10^6; the Leech lattice's $\zeta(s)\zeta(s-11) - L(\Delta, s)$ has no zeros at all on $\Re s = 6$ to $t = 120$, its 57 zeros to height 198 lying near $\Re s \in [11.1, 12.5]$ with spacings more rigid than GUE. The product over primes is lost exactly where Δ enters.

8. Open problems

(1) A conceptual proof of Lemma 4.5 replacing the finite computation at $p = 5, 7$ (the primes 5 and 7 divide some saturation indices without producing isotropic intersections).
(2) The multiplicities of the Arthur-type lifts of Theorem 5.3 against Arthur's multiplicity formula ($a_3 = \pm 81$), and the identification of the Iwahori-new tempered pair through its exceptional theta lift to genus-3 Siegel forms of level 3.
(3) The 32-ions; the Okubo integral orders of [12] and their idempotent geometry against the conic law.
(4) Part III: the J_E half of the F_4 class set at level 5 (88% of the mass found; exact deficits 720799/96768 and 3).

Appendix A. Code and data

Repository `chrisyounge-create/Cayley-Agent-Repository`: `sedenion.py` (doubling, laws), `sedenion_geometry.py`, `sedenion_step2.py`, `sedenion_step3.py`, `sedenion_step3_complex.py` (spectral geometries), `zd_hecke.py` (stabiliser, annihilator, multiplicities), `g2table/zd_periods.py`, `g2table/zd_periods_vec.py` (periods), `g2table/lemma43.py` (proof of Lemma 4.5), `g2table/zd_periods_level`

(level-7 periods and lifts), `beurling.py` (prime-correlation model). Dropbox notes `RH/SEDENIONS.md` through `RH/NEXT_RUN_ITEMS_1_TO_4.md` contain the session records.

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